

Lecture 9: Risk Aversion vs Intertemporal Substitution

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Intro

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Building Blocks

What is Risk Aversion?

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- X : a set of outcomes; $\Delta(X)$: the set of lotteries over X (the simplex)
- Let $\succeq$ be a preference relation over $\Delta(X)$
- You are **risk-averse** if, for any $p \in \Delta(X)$, you have $\delta_{\mathbb{E}[p]} \succeq p$

What Von-Neumann and Morgenstern Taught Us

- Typical assumptions: $\succeq$ is complete, transitive, and continuous
- Independence axiom: if $p \succeq q$, then for any r and $\alpha \in (0, 1)$, we have $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$

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- Under independence, $\succeq$ can be represented by a utility function $U : \Delta(X) \rightarrow \mathbb{R}$ such that

$$U(p) = \sum_{x \in X} \pi(x)u(x)$$

where $\pi(x)$ is the probability of outcome x and $u(x)$ is the utility of outcome x

- The preferences we will cover in this lecture will violate exactly this axiom, stay tuned!

- The **Certainty Equivalent** of $p \in \Delta(X)$ is the amount of money c such that $\delta_c \sim p$

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- John is (weakly) **more risk-averse** than Mary if $CE_J(p) \leq CE_M(p)$ for all $p \in \Delta(X)$
- Pratt showed that this is all about how concave u is!
- Concavity is measured by second derivatives... but these are not scale-free!
- We typically measure risk aversion by $-u''(\cdot)/u'(\cdot)$ or $-x \cdot u''(x)/u'(x)$
- Many prominent utility functions: CARA, CRRA, HARA, ...

The Special Case of CRRA

- In Macro, we often use the CRRA utility function:

$$u(C) = \frac{C^{1-\gamma} - 1}{1-\gamma}$$

- The parameter γ is the coefficient of relative risk aversion (RRA)

$$\frac{-c \cdot u''(c)}{u'(c)} = \gamma$$

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- If $x \succeq y$, then $\alpha x \succeq \alpha y$ for any $\alpha > 0$ (Indifference curves are radially symmetric)
- Optimal portfolio weights are independent of wealth (with complete markets)
- Equilibrium prices (and returns) depend on *growth rates*, not levels

Questions?

Intertemporal Substitution

- You don't need the concept of *uncertainty* to think about intertemporal substitution
- Think about two periods and a consumer choosing optimal consumption:

$$\max_{C_1, C_2} u(C_1) + \beta u(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1+r} = W$$

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- Exercise: show that for CRRA preferences, the EIS is $1/\gamma$

Risk Aversion vs Intertemporal Substitution

- With CRRA preferences, high $\gamma \implies$ high risk aversion
- But high $\gamma \implies$ low EIS $\implies$ low willingness to substitute consumption
- Only two possibilities:
 - High γ : very risk-averse, hates shifting consumption around
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 - Low γ : not very risk-averse, and very willing to shift consumption around
- If you bump up γ to make the consumer afraid of equity...
- She will really upset about shifting consumption around!
- You have to make the relative price of consuming tomorrow low $\implies$ high risk-free rate!
- The prize for waiting must be large... risk-free rate puzzle from Weil (1989)!

Questions?

- Campbell, Chap. 6

Weil, Philippe. 1989. "The Equity Premium Puzzle and the Risk-Free Rate Puzzle." *Journal of Monetary Economics* 24 (3): 401–21. [https://doi.org/10.1016/0304-3932\(89\)90028-7](https://doi.org/10.1016/0304-3932(89)90028-7).